UNIT CIRCLE AND SPECIAL TRIANGLES

PLANE TRIGONOMETRY

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Unit Circle

$(\cos\theta, \sin\theta)$ coordinate

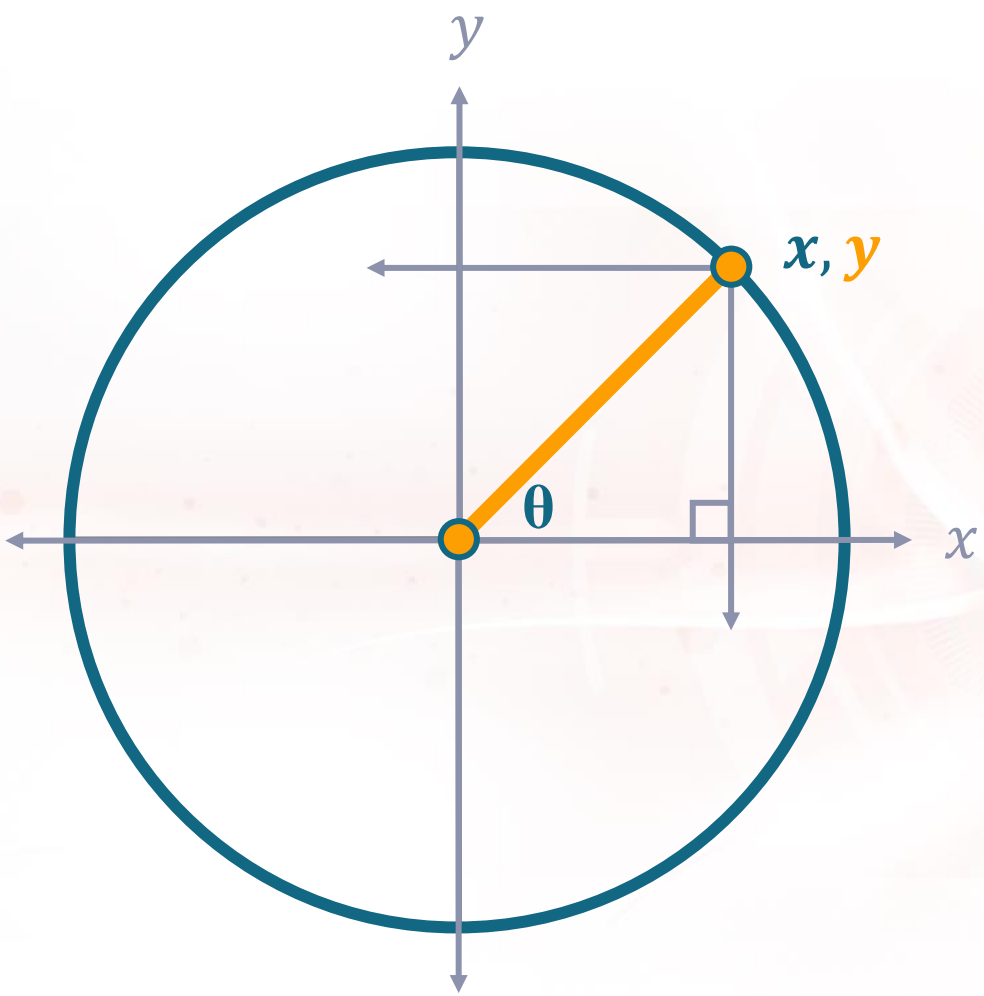
Special Right Triangle

45-45-90

30-60-90

UNIT CIRCLE

UNIT CIRCLE



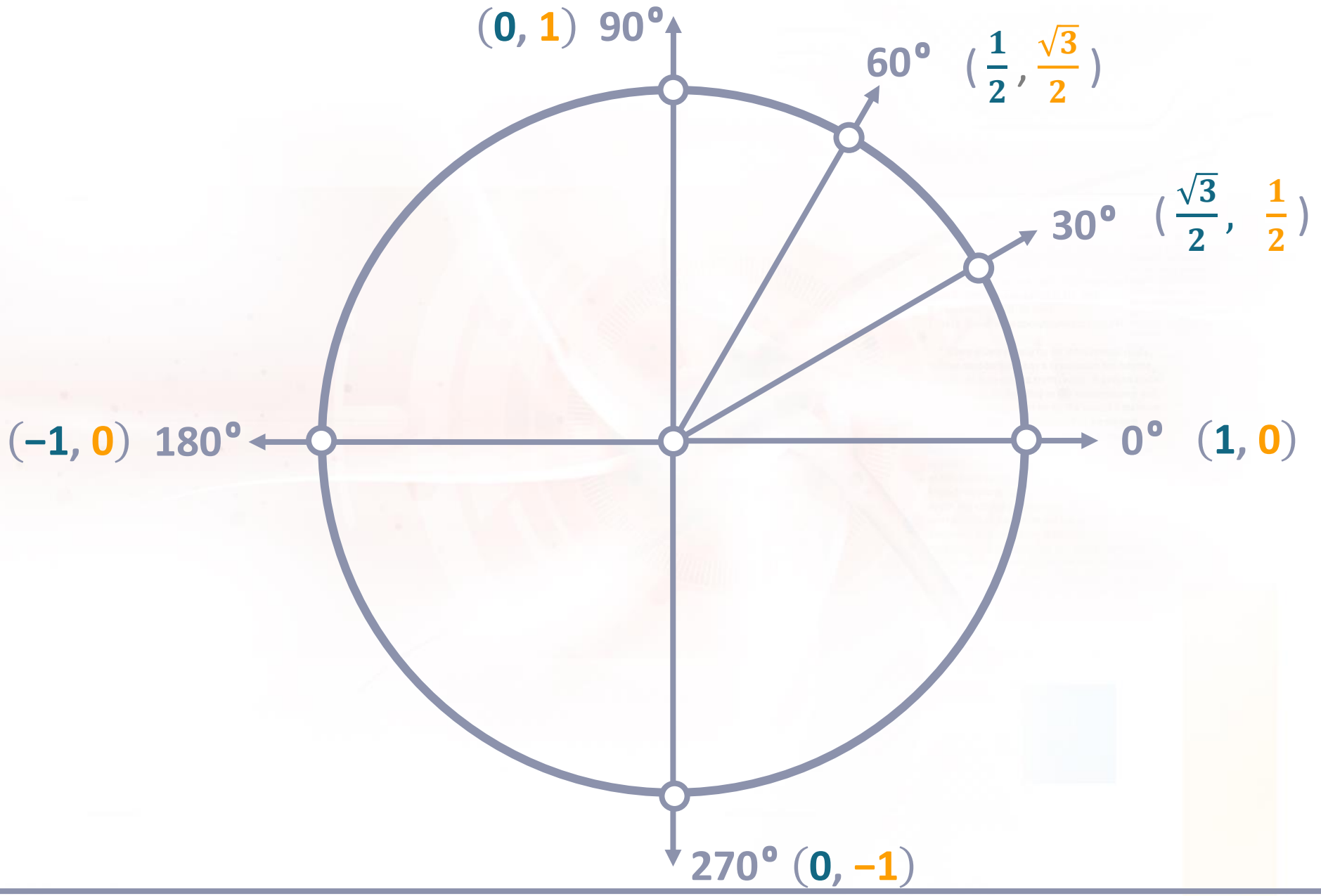
A circle with a **radius** of **1-unit**

$$\sin\theta = \frac{y}{1}$$

$$\sin\theta = y$$

$$\cos\theta = \frac{x}{1}$$

$$\cos\theta = x$$


(0, 1) 90°
60° $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
30° $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$
(−1, 0) 180°
0° (1, 0)
270° (0, −1)

Show that,

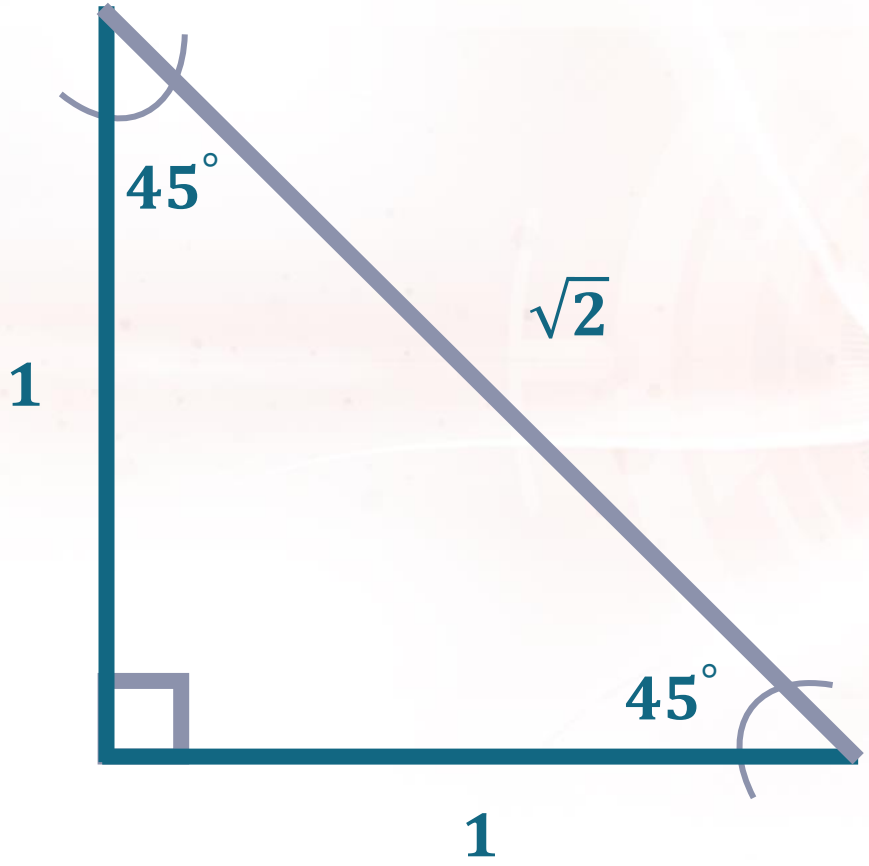
$$\cos 60^\circ = 2\left(\cos 30^\circ\right)^2 - 1$$

Show that,

$$5\tan^2 30^\circ = \sec 60^\circ - \frac{\csc 90^\circ}{\csc 30^\circ + 1}\cos 0^\circ$$

SPECIAL RIGHT TRIANGLES

45-45-90 TRIANGLE

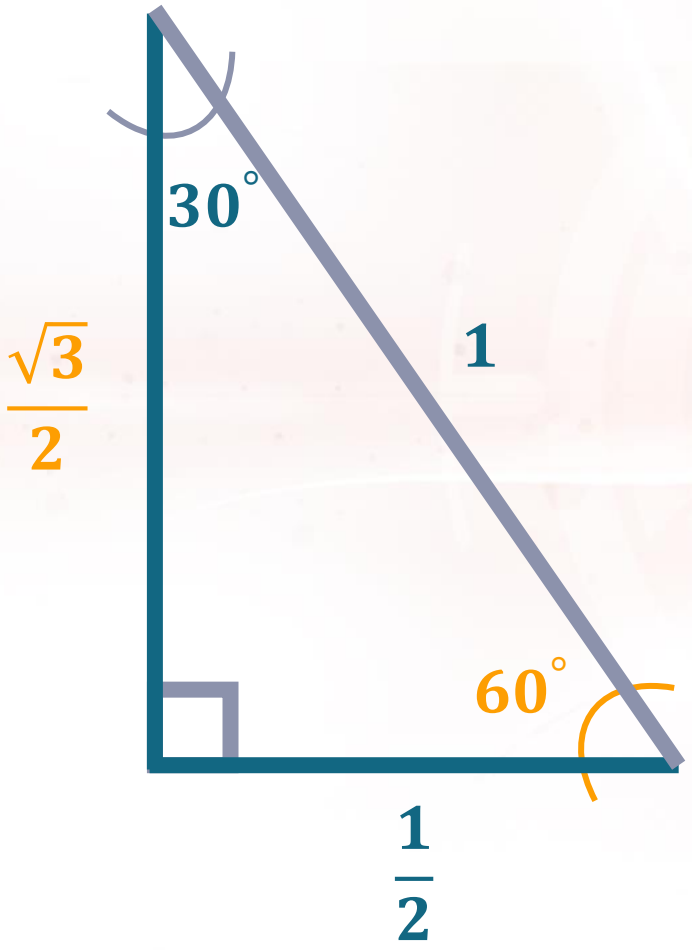


$$\sin 45^\circ = \frac{1}{\sqrt{2}}$$

$$\cos 45^\circ = \frac{1}{\sqrt{2}}$$

$$\tan 45^\circ = 1$$

30-60-90 TRIANGLE



$$\sin 30^\circ = \frac{1}{2}$$

$$\sin 60^\circ = \frac{\sqrt{3}}{2}$$

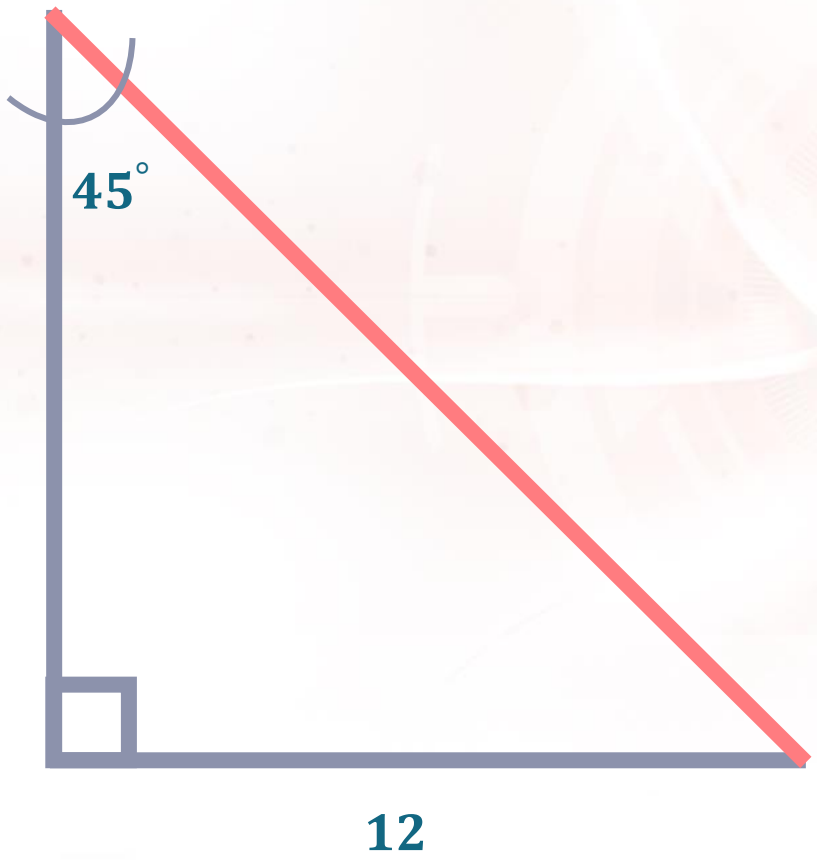
$$\cos 30^\circ = \frac{\sqrt{3}}{2}$$

$$\cos 60^\circ = \frac{1}{2}$$

$$\tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\tan 60^\circ = \sqrt{3}$$

In the triangle below, what is the length of **hypotenuse** ?



In the triangle below, what is the length of each **side** ?

SEATWORK